

Group, Task, and Personality Correlates of Boundary-Spanning Activities

Robert C. Dailey¹

University of Denver

Boundary-spanning activities were studied in 15 organizations engaged in basic and applied research. Included in the study were 281 scientists and engineers. Contrary to prior theory and research, this study found boundary-spanning activities to be unrelated to job satisfaction. It was strongly related to perceptions of research and development team collaboration, job motivation, task uncertainty, locus of control, team cohesiveness, and individual productivity. The research reported here makes a strong case for including group processes and characteristics in future studies involving boundary-spanning activities. The results also give increased impetus to research which examines the relationships between boundary-spanning activities and individual productivity.

INTRODUCTION

A variety of researchers and theorists have pointed out the importance of boundary-spanning activities (Keller, Szilagyi, & Holland, 1976; Thompson, 1967; Aldrich, 1971) at both the group and organizational levels of analysis. At the organizational level, the theme in these empirical studies and theoretical elaborations has been that boundary-spanning activities reduce uncertainty and clarify paths leading to desired goals for organizations. Thus, organizations and their components try to become more proficient at reducing and programming uncertainty so that standardized response patterns can be developed.

¹Requests for reprints should be sent to Robert C. Dailey, College of Business Administration, University of Denver, Denver, Colorado 80208.

Boundary-spanning activities (BSA) are important processes in the continued organizational search for reducing and controlling uncertainty. The purpose of the present research study was to study the relationships between BSA and scientists' and engineers' perceptions of the following variables: (1) group collaboration, (2) team size and cohesiveness, (3) task uncertainty, (4) affective job responses (motivation and satisfaction), and (5) locus of control. The relationship between BSA and individual productivity was also studied. These relationships were investigated in a heterogeneous sample of private and public research and development organizations.

LITERATURE REVIEW

Keller et al. (1976) state that BSA is "the interpersonal transfer of information across organizational boundaries." They further state in their literature review that BSA are vital for maintaining effective organizational functioning and viability. Thus, these researchers view BSA as a critical process for insuring organizational survival.

At the individual level of analysis, Keller et al. (1976) found that BSA were unrelated to role conflict and ambiguity and positively related to job satisfaction for a sample of 192 managers, engineers, and supervisors. These researchers also found that managers and engineers positively related BSA to job satisfaction and job characteristics (variety, significance, identity, friendship, dealing with others, and autonomy).

The task characteristics of interdependence and uncertainty have been shown to be related to individual perceptions of group processes (Van De Ven, Delbecq, & Koenig (1976). Van De Ven et al. (1976) have shown that both are positively related to personalized and group-oriented coordination patterns for employment security agency groups. This finding confirmed Thompson's additive hypothesis (1967). Thus, as task uncertainty and interdependence increase, all coordination modes increased in frequency of use, especially mutually adjustive coordination. At present, little research has been conducted on the relationship between task uncertainty and BSA. It would seem logical to conclude that as task uncertainty increases, BSA would increase due to rising levels of task-related tension and anxiety associated with increases in the probability of task failure.

The Keller et al. (1976) study also found that first-level supervisors positively related BSA to role conflict. They also perceived a negative relationship between BSA and job satisfaction with opportunities for promotion. These findings run counter to those of Kahn, Wolfe, Quinn, Snoek, and Rosenthal (1964). These researchers found in their national

survey that individuals occupying positions requiring a high level of BSA experienced greater levels of role conflict, tension, and feelings of being misunderstood by role senders. Organ and Greene (1972) found that boundary-spanning managers perceived lower role consensus and compliance with their superiors than those managers low on this job dimension. Similar conclusions have been reached in experiments conducted by Wall and Adams (1974) and Organ (1971). These studies taken together point out the consistent relationships between role perceptions and BSA. However, they do not specify the effects of perceptions of group processes on BSA.

It has also been shown that role conflict and ambiguity are related to locus of control. For instance, Organ and Greene (1974) and Szilagyi, Sims, and Keller (1976) have shown that locus of control is positively related to role conflict and role ambiguity. Thus, as the individual's assessment of reward causality becomes more external, perceived ambiguity and conflict increase. Ruble (1976) found that locus of control is negatively related to group productivity under conditions of participation. Thus, it may be concluded that the personality dimension locus of control is related to tolerance for role ambiguity, role conflict, and certain types of performance. It may also be that locus of control is negatively related to BSA, e.g., as the individuals' orientations become more internal, their levels of performance or productivity may increase.

The above research studies indicate the importance of job characteristics and personality for understanding BSA. None of the studies, however, attempt to assess the effects of group characteristics or processes along with task characteristics on perceptions of BSA. None of these research studies have specifically assessed the impact of BSA on individual productivity.

Some tangentially related group process research has been conducted by Aram, Morgan, and Esbeck (1971) in the area of research and development (R&D) team processes. For their sample of R&D personnel, it was found that knowledge-based risk taking (a factor-analytic dimension of collaboration) was related to individual productivity and affective reactions to work (satisfaction with professionalism and opportunities for self-actualization). Regarding the research proposed here, it would seem possible that those individuals who are more risk-oriented regarding group tasks may also be boundary spanners for the group. Furthermore, knowledge-based risk taking may help clarify roles in work groups (reduce ambiguity) and thus increase the overall level of BSA in the group. The notion of role clarification is evident in Barth's study (1971) of R&D personnel, in which he found that risk taking was negatively related to the level of communication and coordination problems in R&D teams. Taylor and Utterbeck (1975) found that over time the level of technical communications between mem-

bers of different R&D teams decreased if changes in administrative control (alteration of team composition) had occurred. This relationship did not occur if the task or project was changed, however. Rubenstein, Barth, and Douds (1971) found that the level of communication problems between R&D teams was negatively related to intergroup dependence and intergroup climate. It is likely that clarification of roles for professionals in these studies may have increased their BSA.

Taken together, the above studies indicate that team processes (collaboration and coordination) may affect the conditions under which BSA occur. Thus, if R&D team members perceive their teams to be collaborative and coordinated in a personalized (nonbureaucratic) manner, then it is more likely that their BSA will increase, especially if these processes are to clarify role requirements for work group members.

At the present time little research exists which addresses the relationships between group characteristics (size and cohesiveness) and BSA. Group size and cohesiveness have been studied as antecedents of a variety of group and individual outcomes (Cartwright, 1968; Steiner, 1966; 1972). For instance, Cartwright (1968) notes that cohesiveness is related to levels of member anxiety, stress, perceived group attractiveness, and group atmosphere. Similar results are evident regarding the effects of group size, although most of the relationships are in the opposite direction (Hackman & Vidmar, 1970; Steiner, 1972; Kahn et al., 1964). Regarding the theory being advanced here, for highly professionalized work groups, as the groups become smaller and more cohesive BSA may increase among members.

In summary, considerable evidence has been generated which indicates that BSA are related to role perceptions, task characteristics, and personality characteristics. Little research has been conducted which studied the relationship between perceptions of group processes and BSA. Few studies have been conducted which have assessed the relationship between group characteristics (size and cohesiveness) and BSA. Finally, it is not presently known how these variables interact with BSA to determine individual productivity.

Given the theoretical and empirical state of affairs noted above, several hypotheses and research questions emerge. Hypotheses are formulated if a body of research exists to support the assumed relationship among the variables. Research questions are formulated if the assumed relationship among the variables has not been addressed in the literature. The hypotheses and research questions developed from the literature review are presented below.

Hypothesis 1: BSA will be negatively related to locus of control.

Hypothesis 2: BSA will be positively related to job satisfaction.

Hypothesis 3: BSA will be positively related to individual motivation.

Research question 1: BSA will be positively related to perceived group cohesiveness and collaboration.

Research question 2: BSA will be negatively related to task uncertainty and team size.

Research question 3: BSA will be positively related to individual productivity.

METHOD

Sample

The study was conducted in 15 R&D organizations located in the Rocky Mountain west. R&D activities in these organizations ranged from environmental research to defense contracting and space exploration. The subjects were 281 scientists and engineers who were organized into 45 project groups ranging in size from 2 to 19 members. The average age of the subjects was 38.2 years; the average time in the organization was 8.55 years; 98% of the subjects were college graduates. Finally, 26% of the respondents held PhD degrees and 18% held master's degrees.

Questionnaires were distributed to project leaders or team leaders by the researcher. These leaders then distributed questionnaires to individual team members. Project leaders also were asked to rate the performance of individual team members. All respondents and organizations were guaranteed confidentiality of information.

Measures

Boundary-Spanning Activities (BSA). A four-item BSA scale based on the work of Keller et al. (1976) was used to assess boundary spanning in this study. This measure was chosen because it was originally developed by these authors to study BSA in research and development organizations (Keller & Holland, 1975a, 1975b); the range on this construct is zero to 20. Reliabilities are reported in Keller et al. (1976).

Individual Productivity (IP). Productivity of scientists and engineers was obtained by asking their project leaders to assess their contributions to general or scientific knowledge (Farris, 1969; Pelz & Andrews, 1966) and their overall usefulness to the organization (Farris, 1969; Pelz & Andrews, 1966). The first performance dimension was a measure of the professional's effect on the body of knowledge in his or her discipline, while the second measure assessed the professional's contribution to the furtherance of

organizational goals. These two items were combined to form an index of individual productivity, since the dimensions correlated at $r = .837$ ($p \leq .001$) and $r = .829$, ($p \leq .001$) with IP respectively. Scoring for both items was accomplished by using a 10-point scale. The range was therefore zero to 20 for IP.

Locus of Control (LC). The personality construct locus of control was measured with Valencha's shortened version (1972) of Rotter's instrument (1966). This scale measures personal beliefs about events (reinforcements) and whether they are under the control of the individual or his or her environment (Rotter, 1966). The theoretical range of scores for this construct is 10 to 44, with 10 being maximum internal orientation. Reliabilities for the shortened instrument are reported in Valencha (1972).

Perceived Team Collaboration (PTC). Collaboration was measured with an 18-item instrument developed by Aram et al. (1971). The instrument was designed to measure "the presence of mutual influence between team members, open direct communication and conflict resolution, and support for innovation and experimentation" (Aram et al., 1971). Factor analysis using a Varimax rotation with a Kaiser normalization of the collaboration instrument produced two factors: (1) open, supportive problem solving (COLLAB1), and (2) personalized responsibility in problem solving (COLLAB2) (Dailey, 1977). For COLLAB1 and COLLAB2 the correlations between each factor and the total collaboration (COLLABT) score were .892 ($p \leq .001$) and .818 ($p \leq .001$) respectively. The scoring range for COLLAB1 is zero to 16.6; for COLLAB2 it is zero to 11.2. Thus, the range for COLLABT is zero to 27.8. The reason these ranges have variable intervals is the fact that item-factor loadings were used as weights to convert respondent's raw scores to factor scores. A similar explanation also applies to the cohesiveness construct which was also factor analyzed. The reliability coefficients for internal consistency of collaboration factors using Kuder-Richardson formula 20 were .99, .92, and .99 respectively.

Team Size (TS). The size of the R&D team was determined by the number of questionnaires assigned to each project leader. If not all team members completed the instrument, TS was adjusted to reflect the attrition.

Individual Job Satisfaction (IJS). Job satisfaction was measured in this research with the Job Diagnostic Survey (JDS) (Hackman & Oldham, 1974). IJS is defined by Hackman and Oldham (1974) as "the degree to which the employee is satisfied and happy in his or her work." The range on this variable is $-.6$ to 5.8 . Reliabilities for IJS are reported in Hackman and Oldham (1974).

Individual Motivation (IM). Motivation was also measured with the JDS (Hackman & Oldham, 1974). It is defined as "the degree to which the employee is self-motivated to perform effectively on the job." The range on this variable is $.3$ to 6.0 . Reliabilities for IM are reported in Hackman and Oldham (1974).

Task Uncertainty (TU). Task uncertainty was measured using an eight-item instrument developed by Van De Ven et al., (1976). A seven-point Likert format was used for each item. According to the authors, TU is "the difficulty and variability of work undertaken by an organizational unit." Regarding scoring, maximum TU equaled zero, while minimum TU equaled 50. Reliabilities for the construct are presented in Van De Ven et al. (1976).

Cohesiveness (COH). Cohesiveness was measured in this study by a 19-item questionnaire that was factor analyzed in the same way as the collaboration instrument. Items making up this construct were taken from literature reviews written by Cartwright (1968), Stogdill (1972), and Pelz and Andrews (1966). The factor analysis produced two factors: (1) team attractiveness (COH1), and (2) involvement and satisfaction with the team (COH2). Specific structures of these factors are reported in Dailey (1977). The range for COH1 is 6.24 to 29.66; for COH2 is 2.68 to 13.37. For COHT the range is 8.92 to 43.03. For COH1 and COH2 the correlations between each factor and the total cohesiveness (COHT) score were .926 ($p \leq .001$) and .874 ($p \leq .001$) respectively. The reliability coefficients for internal consistency of factors using the Kuder-Richardson formula 20 were .98, .98, and .99 for the total score.

RESULTS

The correlational analysis performed on the sample ($N = 281$) is reported in Table I. The analysis shows strong relationships between BSA and collaboration, individual motivation, task uncertainty, locus of control, and all dimensions of perceived team cohesiveness. Also, the analysis for BSA and individual productivity produced a significantly positive correlation ($r = .237, p \leq .001$).

From Table I it is apparent that BSA is negatively related to locus of control ($r = -.233, p \leq .001$). Thus, as locus of control becomes more internalized (smaller scores), BSA increase. Thus, hypothesis 1 is strongly supported by the results of this research. Hypothesis 2 was not supported, since BSA is not positively related to IJS. However, BSA is positively related to individual motivation (IM) ($r = .201, p \leq .001$); thus hypothesis 3 is supported.

Research question 1 stated that a positive relationship should be observed between BSA and perceptions of team cohesiveness and collaboration. Table I indicates that respondents perceive open supportive problem solving (COLLAB1), personalized responsibility in problem solving (COLLAB2), and total collaboration (COLLABT) to be all positively related to BSA. In a similar fashion, team attractiveness (COH1), involvement and satisfaction with the team (COH2), and total cohesiveness

Table I. Pearson Product Moment Correlations Between Predictor and Boundary-Spanning Activity (BSA) and Individual Productivity (IP)

Variable	BSA
COLLAB1	.287 ^a
COLLAB2	.334 ^a
COLLABT	.358 ^a
TS	-.032
IJS	.052
IM	.201 ^a
TU	-.276 ^a
LC	-.233 ^a
COH1	.370 ^a
COH2	.378 ^a
COHT	.413 ^a
IP	.237 ^a

^a $p \leq .001$.

(COHT) are all positively related to BSA (average $r = .387$, $p \leq .001$). These results therefore, strongly support research question 1. BSA are indeed related to team collaboration and cohesiveness.

Research question 2 stated that a negative relationship between BSA and task uncertainty (TU) as well as team size (TS) should exist. This research question is only partially supported, since only the correlation for TU was significantly negative ($r = -.276$, $p \leq .001$). The question was not supported for TS, although the results were in the proper direction. It should be recalled that as TU increases (smaller scores) BSA increase. Thus, respondents in this sample are more likely to engage in BSA as the task they confront becomes more uncertain.

Research question 3 stated that a positive relationship would be observed between BSA and IP. This correlation was noted earlier, and its value was .237 ($p \leq .001$). Thus, question 3 is strongly supported in these sample results.

Tables II and III present the results of two stepwise multiple regression analyses. Table II illuminates the predictive relationship between the independent predictors and BSA. Table III includes BSA as a predictor of individual productivity (IP). Since the sample was nearly equally divided between publicly and privately employed professionals, it was deemed necessary to see if this categorical variable, e.g., subsample membership, was having an impact on the predictive power of the regression models for either BSA or IP. This required that analysis of covariance

Table II. Results of Stepwise Multiple Regression Using Study Variables to Predict BSA ($n = 281$)

Variable	<i>B</i>	<i>F</i>	<i>R</i> ²	ΔR^2
COHT	.162	57.22 ^a	.170	.170
TU	-.299	16.15 ^a	.216	.046
LC	-.116	5.23 ^b	.250	.034
COLLAB2	.275	3.39 ^b	.264	.014
IM	-.847	3.78 ^b	.282	.018
IJS	-.441	1.55	.290	.008
Constant value	3.33	5.10 ^c		

^a $p \leq .001$.^b $p \leq .01$.^c $p \leq .05$.

(ACOVA) be performed with sector membership as a dummy variable in the predictor set (Nie, Hull, Jenkins, Steinbrenner, & Bent, 1975).

The homogeneity test (Nie et al., 1975) for BSA produced an *F* ratio equal to .831 with 16 and 264 degrees of freedom. Therefore it may be concluded that the inclusion of sector membership as a predictor in the regression model did not significantly increase the predictive power of the model. For this reason, Table II presents only the results of using the metric independent variables to predict BSA. A similar analysis (ACOVA) was undertaken for IP with BSA included in the predictor set. Here the homogeneity test produced an *F* ratio equal to 1.34 with 18 and 246 degrees of freedom.

Table III. Results of Stepwise Multiple Regression Using BSA and Study Variables to Predict Individual Productivity (IP) ($n = 281$)

Variable	<i>B</i>	<i>F</i>	<i>R</i> ²	ΔR^2
LC	-2.21	26.50 ^a	.087	.087
TS	-2.06	10.96 ^a	.122	.035
BSA	1.31	9.66 ^b	.151	.030
COH1	1.43	3.83 ^c	.163	.012
IJS	2.02	1.99	.169	.006
TU	1.28	1.98	.175	.006
COLLAB2	1.96	1.37	.180	.005
COH2	-1.33	1.50	.188	.008
Constant value	118.48	17.11 ^a		

^a $p \leq .001$.^b $p \leq .01$.^c $p \leq .05$.

Once again, this F ratio was not significant; therefore only the metric predictors are reported in Table III.

In Tables II and III the following items are reported: (1) the standardized beta coefficients in the order in which they entered the regression model; (2) the F ratio and its level of significance for each beta coefficient; (3) the multiple R -squared, and (4) the change in R -squared associated with each beta coefficient. The Y intercept (constant) and its significance level is also reported.

Table II indicates that, when considering BSA, aspects of all the predictors with the exceptions of team size (TS) and individual job satisfaction (IJS) were significant predictors of BSA. COHT was the strongest predictor, since it entered the model first and explained 17% of the variation in BSA. It should be noted that TU possesses a negative coefficient; thus as TU increases (smaller scores) BSA increase. A similar pattern is also evident for locus of control (LC). As the professionals' orientations became more internal, BSA were likely to increase.

Scientists and engineers who perceived personalized responsibility in problem solving (COLLAB2) were also more likely to pursue BSA ($F = 3.39, p \leq .05$). Finally, those professionals who demonstrated higher levels of individual motivation (IM) were also more likely to demonstrate BSA. Individual job satisfaction (IJS) was not a significant predictor of BSA. It should be noted, however, that the IJS beta coefficient is negative.

The overall R -squared for this regression model was .290 ($F = 7.96, p \leq .001$). Therefore, it can be concluded that aspects of professionals' personalities (LC), perceptions of team processes (COLLAB2), task characteristics (TU), and affective job responses (IM) are significant predictors of BSA.

Table III presents a similar analysis using individual productivity (IP) as the criterion, with BSA included in the set of predictors. In this analysis, four of the 12 predictors possessed significant beta coefficients (LC, TS, BSA, and COH1). Both LC and TS have negative coefficients; therefore as professionals' internal orientation increased and as team sizes became smaller, IP increased. A number of other predictors entered the model; however, their beta coefficients were not significant (see Table III).

BSA was a significant predictor of IP ($B = 1.31, F = 9.66, p \leq .01$). Thus, more productive scientists and engineers were likely to be obtaining technical information from sources outside their organizations and their R&D teams. Finally, professionals perceived a link between team attractiveness (COH1) and IP ($B = 1.43, F = 3.83, p \leq .05$). Therefore, as perceived team attractiveness increased, so did IP. Overall, the R -squared for the regression model of IP was .188 ($F = 4.65, p \leq .001$).

DISCUSSION

The results of this study are basically supportive of the overall importance of team collaboration, team characteristics, and task uncertainty for understanding BSA and IP in the R&D setting. However, the predictor set had greater predictive power relative to BSA than IP ($R^2 = .290$, $R^2 = .188$ respectively). Aspects of team cohesiveness and the locus of control construct were important in both models. For BSA, motivation, task uncertainty, and collaboration played significant predictive roles, while they were less important regarding IP. For IP, BSA and TS became important predictors.

The fact that the study variables do not predict BSA and IP in the same way implies that scientists and engineers in this sample do not perceive the two activities in the same way. However, they did perceive BSA to be important for enhancing IP, since a significant positive relationship was observed between the two variables. For IP, it appears that BSA and small team sizes enhance one's contributions to the field and the organization.

Scientists and engineers in this sample did relate orientations towards reward causality and team cohesiveness to both criteria. This in itself is an important finding for the area of R&D management. This conclusion tends to reinforce the importance of selection practices for forming teams, and leadership practices which encourage team cohesion and collaboration once teams are established. Therefore it may be wise for the R&D manager to form teams that are more homogeneously internal in their orientations towards reward causality.

The results of this research do not support the findings of Keller et al. (1976) regarding job satisfaction and BSA. They found that positions with high levels of BSA are desirable and more satisfying for incumbents. This finding was not confirmed in the intensive technologies (Thompson, 1967) of R&D organizations. The results reported in these R&D data suggest that job motivation rather than satisfaction is more important for understanding BSA. It would be premature to assume that IJS has no relationship with BSA, since the two studies involved substantially different samples. It is possible that the findings of Keller et al. (1976) are sample-specific, since only one organization was involved in their research.

The findings reported here tend to reinforce the conclusions of Keller et al. (1976) regarding the importance of task characteristics for understanding BSA. This is important, since the two samples are quite different from each other (both with respect to technologies employed and the number of organizations participating in the research efforts). Regarding task uncertainty, this study agrees with the conclusion of Keller et al. (1976) that

“much of the prior theory and research relating to BSA has focused on its negative aspects without fully appreciating the potentially positive aspects that BSA can have for those who engage in this activity.”

The results presented here clearly point out the importance of team characteristics and processes for understanding BSA and IP to a lesser degree. For BSA, it may be that cohesive teams with collaborative problem-solving processes provide a stable base from which team members can become more inquiring about technical aspects of their work without increased levels of anxiety or work-related tension. This is in keeping with Cartwright's argument (1968) that cohesive groups buffer individual members from perceived ambiguity and help reduce anxiety and tension among members. These stabilizing team effects may then allow the R&D team members in this study to deal with perceived task uncertainty in a more creative and innovative way.

The research presented here also points out the importance of studying the relationship between BSA and individual performance. Similar research should be conducted which assesses the relationship between BSA and work group productivity. This researcher would once again agree with Keller et al. (1976) in their expressed desire to see more research on task characteristics and job level as determinants of BSA. However, the research reported here would also encourage the inclusion of group characteristics, group processes, and personality factors in such research designs.

REFERENCES

- ALDRICH, H. Organizational boundaries and interorganizational conflict. *Human Relations*, 1971, 24, 279-293.
- ARAM, J. D., MORGAN, C. P., & ESBECK, E. S. Relation of collaborative interpersonal relationships to individual satisfaction and organizational performance. *Administrative Science Quarterly*, 1971, 16, 289-296.
- BARTH, R. T. Intergroup climate characteristics, perceived communication problems, and unity of effort achieved by task interdependent R&D groups. *Proceedings of the Academy of Management*, 1971, 31, 250-254.
- CARTWRIGHT, D. Nature of group cohesiveness. In D. Cartwright and A. Zander (Eds.), *Group dynamics*. New York: Harper and Row, 1968.
- DAILEY, R. C. *Research and development effectiveness at the individual and team levels: An empirical study*. Unpublished doctoral dissertation, University of Colorado, Graduate School of Business Administration, Boulder, Colorado, 1977.
- FARRIS, G. F. Organizational factors and individual performance. *Journal of Applied Psychology*, 1969, 53, 87-92.
- HACKMAN, J., & VIDMAR, N. Effects of size and task type on group performance and member reactions. *Sociometry*, 1970, 33, 33-35.
- HACKMAN, J. R., & OLDFHAM, G. The job diagnostic survey: An instrument for the diagnosis of jobs and the evaluation of job redesign projects (Tec. Rept. No. 4). Department of Administrative Services, Yale University, 1974.

- KAHN, R. L., WOLFE, D. M., QUINN, R. P., SNOEK, J. D., & ROSENTHAL, R. A. *Organizational stress: Studies in role conflict and ambiguity*. New York: Wiley, 1964.
- KELLER, R. T., SZILAGYI, A. D., JR., & HOLLAND, W. E. Boundary spanning activity and employee reactions: An empirical study. *Human Relations*, 1976, 29, 699-710.
- KELLER, R. T., & HOLLAND, W. E. Boundary spanning roles in a research and development organization: An empirical investigation. *Academy of Management Journal*, 1975, 18, 388-393. (a)
- KELLER, R. T., & HOLLAND, W. E. Boundary spanning activity and research and development management: A comparative study. *IEEE Transactions on Engineering Management*, 1975, EM-22, 130-133. (b)
- NIE, N. H., HULL, C. H., JENKINS, J. G., STEINBRENNER, K., & BENT, D. H. *Statistical package for the social sciences* (2nd Ed.). New York: McGraw-Hill, 1975.
- ORGAN, D. W. Some variables affecting boundary role behavior. *Sociometry*, 1971, 34, 524-537.
- ORGAN, D. W., & GREENE, C. H. The boundary relevance of the project manager's job. *R & D Management*, 1972, 3, 7-11.
- ORGAN, D. W., & GREENE, C. H. The perceived purposefulness of job behavior: Antecedents and consequences. *Academy of Management Journal*, 1974, 17, 69-77.
- PELZ, D. C., & ANDREWS, F. M. *Scientists in organizations*. New York: Wiley, 1966.
- ROTTER, J. B. Generalized expectancies for internal versus external control of reinforcement. *Psychological Monographs*, 1966, 80(1, Whole No. 609).
- RUBENSTEIN, A. H., BARTH, R. T., & DOUDS, C. F. Ways to improve communications between R&D groups. *Research Management*, 1971, 14, 49-59.
- RUBLE, T. L. Effects of one's locus of control and the opportunity to participate in planning. *Organizational Behavior and Human Performance*, 1976, 16, 63-73.
- STEINER, I. Models for inferring relationships between group size and potential group productivity. *Behavioral Science*, 1966, 11, 273-283.
- STEINER, I. *Group process and productivity*. New York: Academic Press, 1972.
- STOGDILL, R. M. Group productivity, drive, and cohesiveness. *Organizational Behavior and Human Performance*, 1972, 8, 26-43.
- SZILAGYI, A. D., SIMS, H. P., & KELLER, R. T. Role dynamics, locus of control and employee attitudes and behavior. *Academy of Management Journal*, 1976, 19, 259-276.
- TAYLOR, R. L., & UTTERBACK, J. M. A longitudinal study of communication in research: Technical and managerial influences. *IEEE Transactions on Engineering Management*, 1975, EM-22, 80-87.
- THOMPSON, J. D. *Organizations in action*. New York: McGraw-Hill, 1967.
- VALENCHA, G. K. Construct validation of internal-external locus of control of reinforcement related to work-related variables. *Proceedings of the 80th Annual Convention of the American Psychological Association*, 1972, 7, 455-456.
- VAN DE VEN, A. H., DELBECQ, A. L., & KOENIG, R. Determinants of coordinations modes within organizations. *American Sociological Review*, 1976, 41, 322-338.
- WALL, J. A., & ADAMS, J. S. Some variables affecting a constituents' evaluations of and behavior toward a boundary role occupant. *Organizational Behavior and Human Performance*, 1974, 11, 390-408.

BIOGRAPHICAL NOTE

ROBERT C. DAILEY is Assistant Professor of Organizational Behavior in the college of Business Administration, University of Denver. He received a bachelor's degree in forestry at Iowa State University, a master's degree in business administration from the University of Northern Iowa, and a doctoral degree in organizational behavior and social psychology from the University of Colorado. His current research includes boundary-spanning activities and correlates of individual and team productivity in research and development organizations.